



USING TIME SERIES ANALYSIS FOR SALES AND DEMAND FORECASTING

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Problem Overview

The purpose of this analysis is to develop and compare time-series forecasting approaches for predicting book sales, using *The Very Hungry Caterpillar* and *The Alchemist* as two representative titles. Accurate demand forecasting is important for supporting inventory planning and reducing the risks associated with both stock shortages and excess inventory.

The analysis focuses primarily on forecasting the final 32 weeks of weekly sales data for each title. Several forecasting approaches are evaluated, including SARIMA, XGBoost and LSTM, together with sequential and parallel SARIMA–LSTM hybrid models. Forecast accuracy is assessed using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE) and Mean Absolute Percentage Error (MAPE). The analysis also investigates monthly forecasting by aggregating the weekly data and comparing XGBoost and SARIMA over the final eight months.

An important characteristic of the dataset is the presence of gaps in the original weekly observations, particularly during 2020–2021. When the data were resampled to create continuous weekly series, these missing periods were represented by zero sales. Consequently, these periods require cautious interpretation, as they do not necessarily represent genuine periods of zero demand.

The overall objective is therefore to determine which forecasting approach provides the most reliable predictions for each title and frequency, and to translate the modelling results into practical recommendations for sales and inventory planning.

Approach

The sales data were resampled into continuous weekly time series for *The Very Hungry Caterpillar* and *The Alchemist*. Exploratory analysis, seasonal decomposition and autocorrelation analysis were used to identify trends, seasonality and data gaps. The final 32 weeks were held out for testing.

Three forecasting methods were evaluated: SARIMA, XGBoost and LSTM. SARIMA models were selected using Auto ARIMA with annual seasonality. XGBoost used lagged sales as predictors and was tuned using Hyperopt and time-series validation. LSTM models used a 28-week lookback, with KerasTuner used to optimize their architecture. Both models generated 32-week forecasts recursively.

Two hybrid approaches were also tested: a sequential SARIMA–LSTM, where LSTM modelled SARIMA residuals, and a parallel combination, where different SARIMA/LSTM forecast weights were evaluated. Finally, sales were aggregated monthly and XGBoost and SARIMA were compared over an eight-month test period. Forecast performance was assessed using MAE, RMSE and MAPE.

Key Results

As shown in **Figure 1**, both titles exhibit substantial weekly sales fluctuations and recurring peaks, with *The Very Hungry Caterpillar* generally recording higher sales volumes. The unusual zero-sales periods around 2020–2021 correspond to gaps in the original data and should therefore not automatically be interpreted as zero demand.

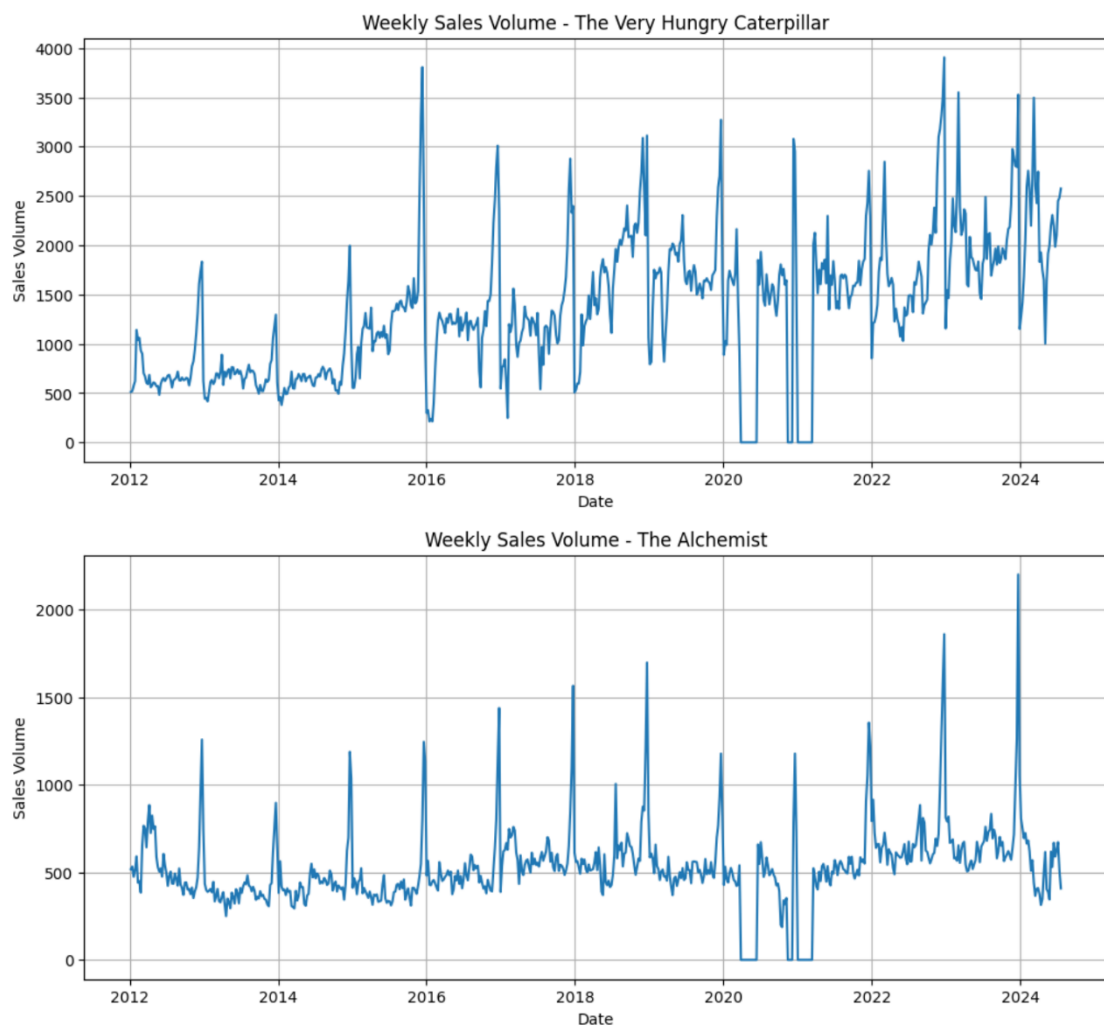


Figure 1. Weekly sales volumes for *The Very Hungry Caterpillar* and *The Alchemist*.

The seasonal decomposition in **Figure 2** identifies long-term trends and clear recurring annual patterns in both titles. This is supported by the autocorrelation analysis, which showed dependence around the 52-week lag and justified incorporating annual seasonality into the forecasting models.

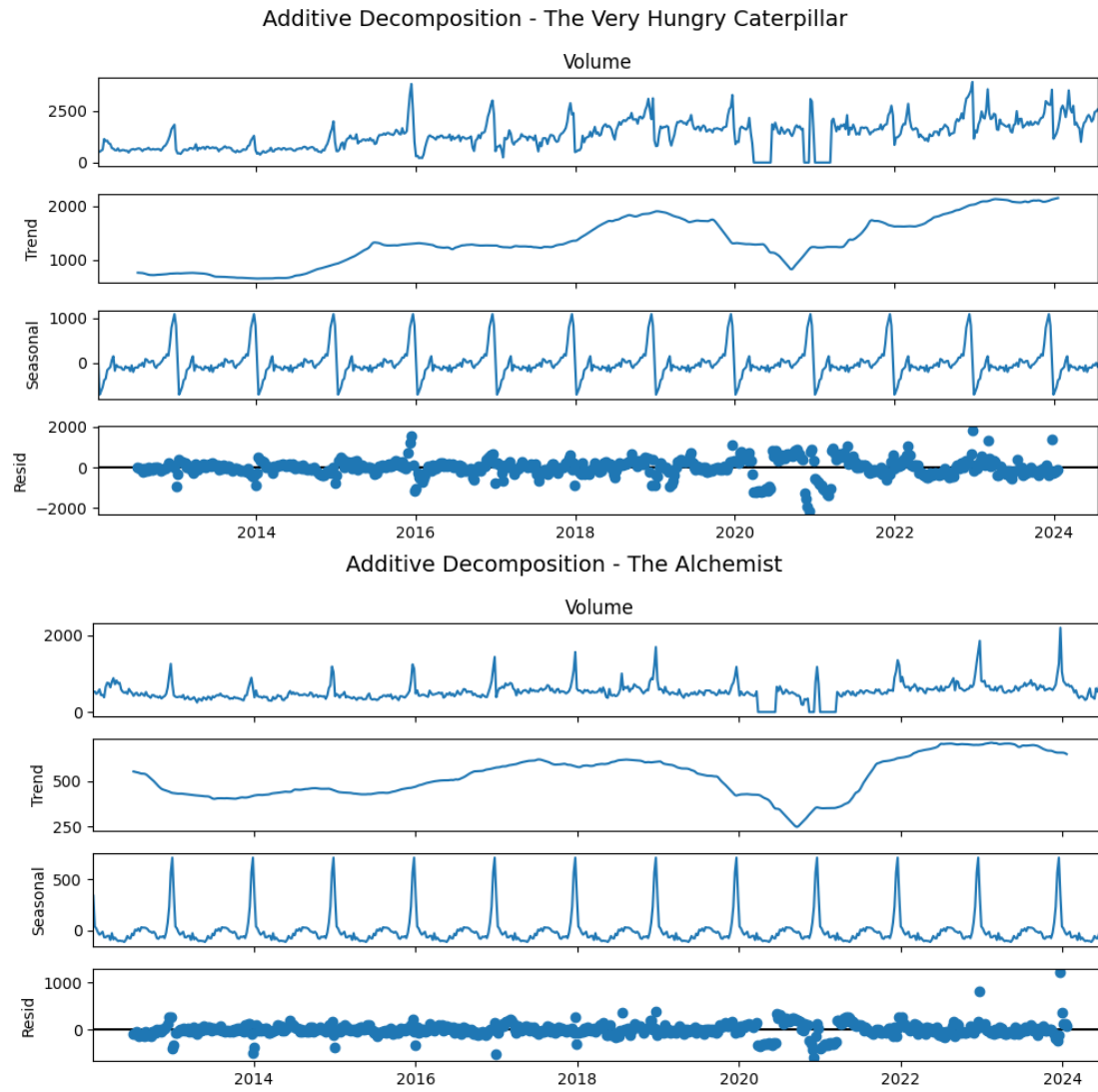


Figure 2. Seasonal decomposition of weekly sales for both titles.

Weekly forecasting results are summarised in **Table 1** for *the Very Hungry Caterpillar*, tuned XGBoost achieved the lowest MAE (341.37) and MAPE (15.76%), while SARIMA achieved the lowest RMSE (448.93). For *The Alchemist*, SARIMA produced the lowest MAE (139.02) and RMSE (213.38), while the sequential hybrid achieved a slightly lower MAPE (22.58%). Parallel combinations did not improve performance, as increasing the LSTM contribution increased forecasting errors.

	The Very Hungry Caterpillar			The Alchemist		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Model						
SARIMA	349.06	448.93	18.47	139.02	213.38	22.92
Tuned XGBoost	341.37	460.76	15.76	159.05	251.33	28.92
Tuned LSTM	460.45	577.73	23.15	209.93	325.29	38.56
Sequential SARIMA-LSTM	355.79	461.10	17.95	140.31	215.10	22.58

Table 1. Weekly forecasting performance over the final 32-week test period.

As illustrated in **Figure 3** and **Figure 4**, the models generally capture the underlying sales level but are less successful at reproducing sudden weekly peaks and declines. For *The Very Hungry Caterpillar* (**Figure 3**), tuned XGBoost follows some short-term movements more closely, consistent with its lower MAE and MAPE, while SARIMA provides a smoother forecast and achieves the lowest RMSE. The LSTM forecast is noticeably smoother and misses several large fluctuations. For *The Alchemist* (**Figure 4**), SARIMA and the sequential hybrid track the observed series closely and produce very similar forecasts, whereas XGBoost and particularly LSTM show larger deviations. This supports the error metrics and indicates that the best forecasting approach differs between the two titles.

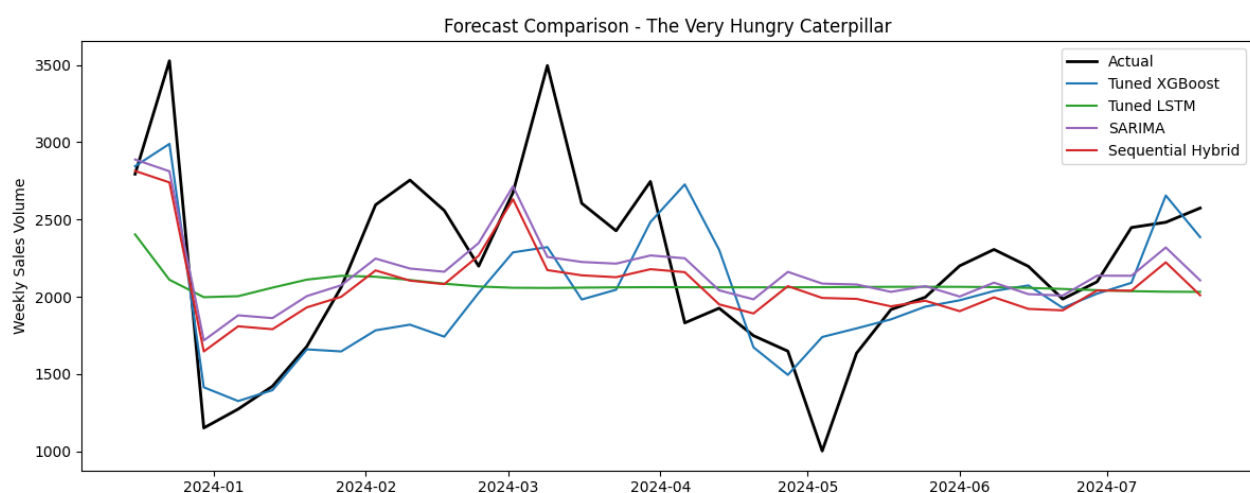


Figure 3. Weekly forecast comparison – *The Very Hungry Caterpillar*.

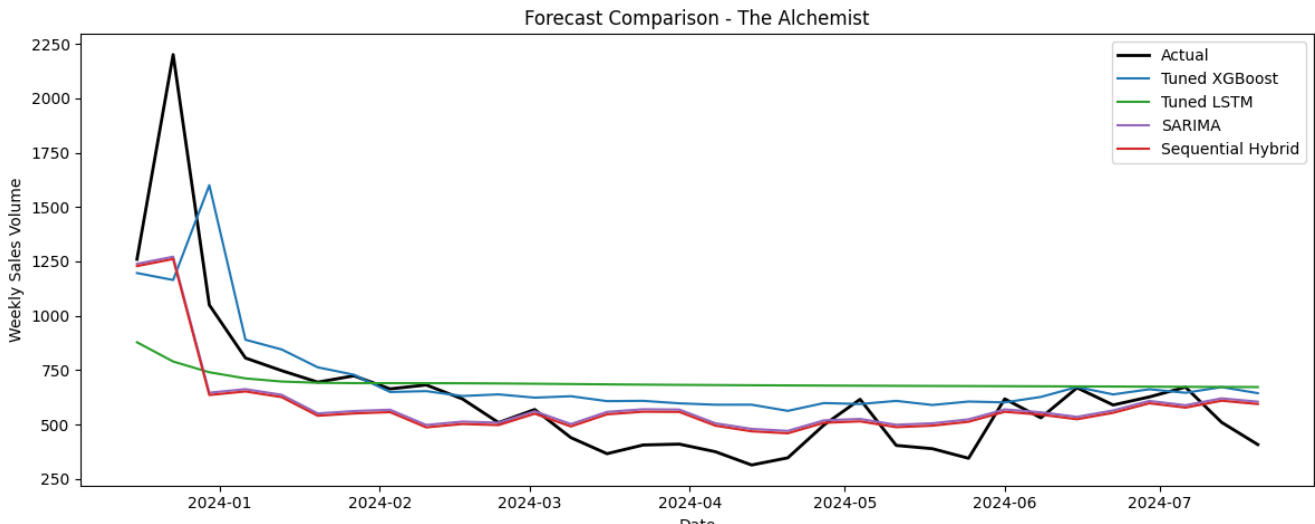


Figure 4. Weekly forecast comparison – The Alchemist.

The monthly results in **Table 2** show that SARIMA outperformed XGBoost across all three error measures for both titles. As illustrated in **Figure 5** and **Figure 6**, both models capture the broad monthly sales level but struggle with larger month-to-month changes. SARIMA generally follows these movements more closely, consistent with its lower forecasting errors. The relatively high monthly MAPE, particularly for *The Alchemist*, also indicates that aggregation to monthly frequency did not necessarily make demand easier to predict.

	The Very Hungry Caterpillar			The Alchemist		
	MAE	RMSE	MAPE	MAE	RMSE	MAPE
Model						
Monthly XGBoost	2574.04	2862.57	26.26	777.81	892.57	37.23
Monthly SARIMA	1955.51	2388.25	19.71	689.79	836.12	30.12

Table 2. Monthly forecasting performance over the final eight-month test period.

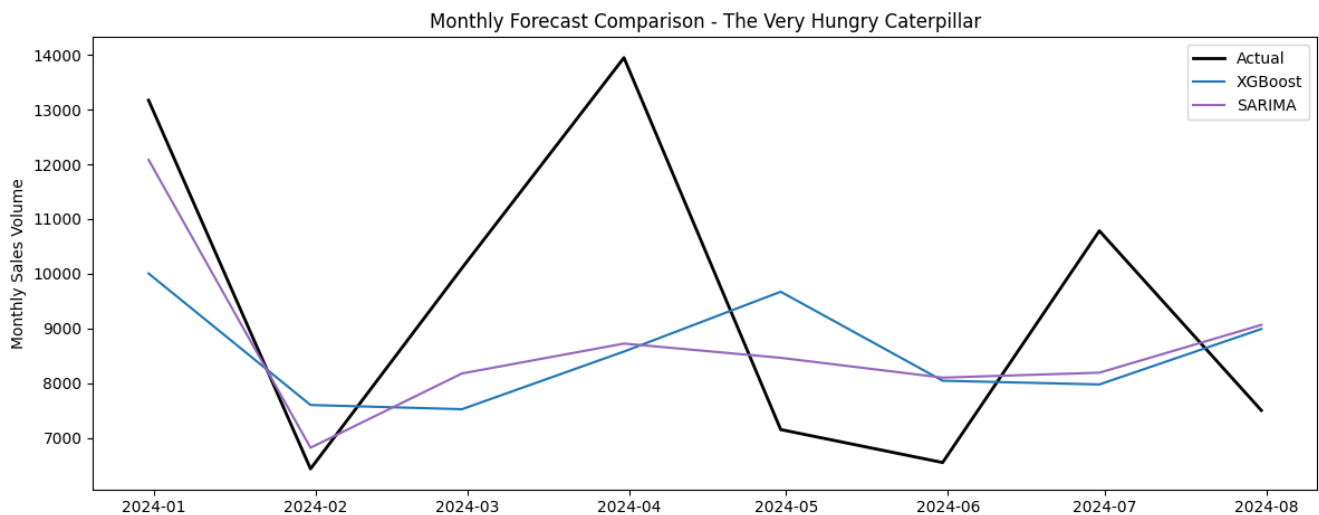


Figure 5. Monthly forecast comparison – The Very Hungry Caterpillar.

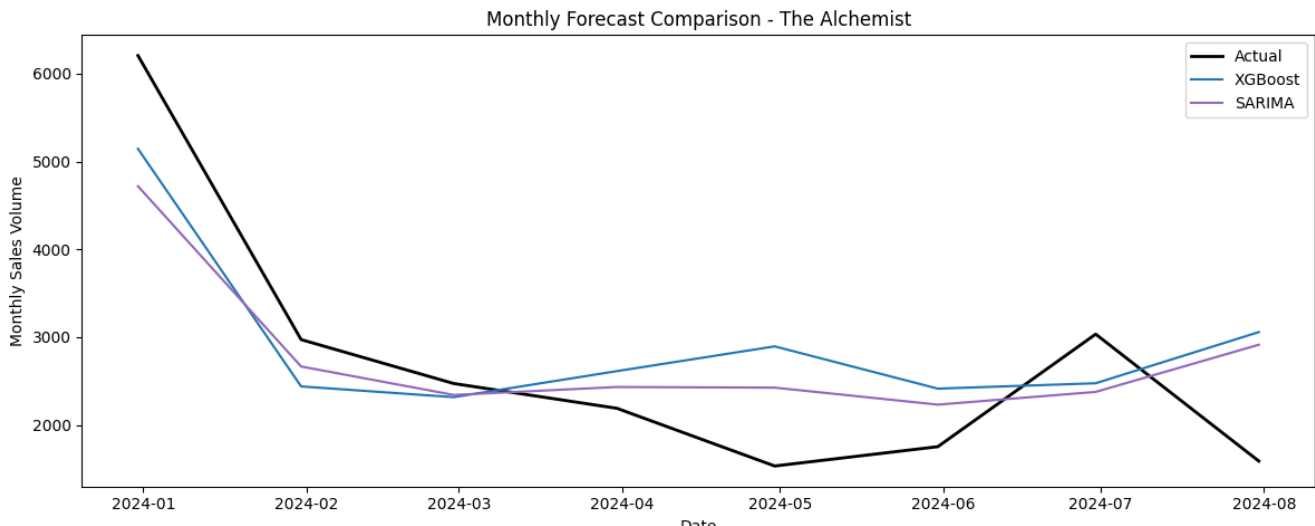


Figure 6. Monthly forecast comparison – *The Alchemist*.

Recommendations

Based on the forecasting results, a title-specific modelling strategy is recommended rather than applying one model to all books. For *The Very Hungry Caterpillar*, tuned XGBoost should be considered for weekly demand planning because it achieved the lowest MAE and MAPE, while SARIMA may be preferred when reducing large forecasting errors is the priority. For *The Alchemist*, SARIMA is recommended because it provides the most consistent weekly performance, with the sequential hybrid offering only a small improvement in MAPE.

The parallel SARIMA–LSTM approach is not recommended, as increasing the LSTM contribution reduced forecast accuracy for both titles. Similarly, the additional complexity of LSTM and sequential hybrid models produced limited benefits relative to simpler alternatives.

For monthly planning, SARIMA should be preferred over XGBoost for both titles. Forecasts should also be reviewed regularly and retrained as new sales data become available, particularly because sudden sales peaks remain difficult for all models to predict.

Conclusion

This analysis demonstrates that forecasting performance varies across both titles and modelling approaches. SARIMA provided the most consistent overall performance, particularly for *The Alchemist* and for monthly forecasting, while tuned XGBoost achieved the strongest weekly MAE and MAPE for *The Very Hungry Caterpillar*. LSTM and hybrid approaches added complexity without consistently improving accuracy. The results also highlight the challenge of predicting sudden sales peaks and the importance of accounting for annual

seasonality and data quality. Overall, selecting models according to individual title behaviour provides a more appropriate forecasting strategy than relying on a single approach for all books.